

PhaseA Summary

Introduction

Cloud security depends on understanding which security tasks are handled by the cloud provider and which ones are the customer's responsibility. Security must be shared while using cloud resources. The AWS Shared Responsibility Model clarifies how it can be shared while using different platforms such as IaaS, PaaS, and SaaS. This model will help in lowering the chances of security incidents that might occur due to poor access control or any misunderstanding related to cloud security.

AWS Shared Responsibility Model

AWS Shared Responsibility Model is responsible for the security of the cloud. This includes physical data centers, networking, hardware, storage infrastructure, and virtualization.

Customer Responsibility

Customers handle security in the cloud. Their responsibilities include managing their data, IAM users and permissions, operating systems, applications, and network configurations.

IaaS vs PaaS vs SaaS:

IaaS :

AWS EC2, Azure VMs

Cloud manages physical hardware, data centers, networking, and virtualization.

Customer manages:

- Operating system patching
- Runtime dependencies
- Applications and data
- Firewalls, IAM policies
- Scaling and load balancing

Analogy

IaaS (Infrastructure as a Service) is like renting an empty office. The building, electricity, and internet connection. We bring our own furniture, computers, and staff. As Infrastructure is already provided, we need to maintain it, keep it running, and pay for the rented Office and utilities. The landlord will be responsible for repairs in regard to the building, whereas we will handle layout, staff, and security of the building.

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PaaS (Platform as a Service)

Examples: Heroku, AWS Elastic Beanstalk, Google App Engine

Provider manages:

- IaaS
- Operating system
- Runtime environment
- Scaling infrastructure

Customer manages:

- Application code
- Application data

Analogy

PaaS (Platform as a Service) is like renting a fully furnished office that will provide furniture, along with computers, and "IT support" - the people who maintain the office infrastructure — like fixing the internet, maintaining the computer, and troubleshooting hardware issues. As infrastructure will be provided (IaaS), along with PaaS services, we just bring our Developers who write/deploy app along with start working on business tasks.

SaaS (Software as a Service)

Example: Gmail

Provider manages:

- Infrastructure(IaaS), platform(PaaS), and the application itself
- Updates, patches, availability

Customer manages:

- Data input and usage
- User access/permissions: who can access
- Configuration of app settings (not the app's code)

Analogy

Suppose we have rented a fully staffed office that will manage the coding of the application, too. We simply use the service while the provider takes care of infrastructure, platform, and software in the form of an application; we only focus on using the application.

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Conclusion

Security comes from sharing responsibility between the provider and its customers; it cannot be handled only by one, even in a SaaS platform. AWS Shared Responsibility Model helps to understand “security of Cloud and security in Cloud” with shared responsibility. With a clear understanding, Cloud Engineers can correctly assign security principles and avoid common misconfigurations. This will help minimize data exposure and unauthorized access.